A Two-Entry Rotor Interface: Period-Four Compression and a Two-Copy Cyclic Decoder

Abstract

We give a worked rotor-routing model with two input vertices, three named outputs, and a branching passive network that feeds back into an active line. For a line of length $k \geq 3$, four complete passive traversals reset all passive rotors. The resulting interface has $3k+4$ canonical states, encoded by a residue z: input at a_j, for $j \in \{k-1, k\}$, adds j modulo $3k+4$. An explicit decoder computes the final phase, internal feedback count, and named output counts for binary input multiplicities. We prove the decoder by a least-stabilization argument and identify its classical sandpile-group interpretation. We then cross-wire two copies at their b_0 outputs and give a polynomial-bit decoder for the four remaining absorbing outputs and both final phases. The physically realized cross count is the least, rather than an arbitrary, algebraic fixed point; a zero-chip example shows why this selection matters. After single-copy compression, polynomial-time batch evaluation also follows from the established Multi-Arrival algorithm for tree-like multigraphs. A separate construction realizes the single interface in a two-sink binary rotor graph whose true shortest-distance levels have width at most three and whose single source run uses both inputs and the feedback edge. The note concerns this restricted interface and its realization, with no general complexity or priority claim.

1 Introduction

A useful rotor interface must retain enough phase information to determine its continuation. A line provides a particularly simple example: on its acyclic rotor states, adding a particle acts as addition in a finite cyclic group. Here the line is coupled to a branching network with a return edge. The passive network has a four-step phase orbit, so the combined system still admits a small exact coordinate. The purpose of this note is to display that coordinate, its output decoder, and the precise scope of the reduction.

The underlying mechanisms are established. Complete rotor stabilization is abelian, chip addition acts on acyclic rotor configurations, and the resulting permutation group is the sandpile group [7, Lemmas 3.9, 3.10 and 3.19]. Least action is part of the general theory of abelian networks [2, Lemma 4.3 and Theorem 4.9]. Moreover, the compressed graph below is tree-like with uniformly bounded contracted height, so the Multi-Arrival theorem of Ghorbani, Hoff and Mnich [4, Theorem 1] already supplies polynomial-time batch evaluation. The explicit single-copy formulas here are a worked specialization of that setting; they do not establish an asymptotic improvement over the existing algorithm. Section 6 uses those formulas to analyze one specified cyclic coupling. Its underlying undirected graph has a four-cycle, so the tree-like theorem does not directly cover the coupled graph.

There are three graph settings in the discussion. The *upper graph* has three absorbing outputs a_0, b_0, c_0 and receives particles at a_k and a_{k-1}. The *coupled graph* cross-wires two copies of that upper interface and has four absorbing outputs. The *physical graph* of Section 8 makes the single interface's three outputs ordinary vertices, adds a lower counter, and has exactly two original sinks.

Its distance levels are measured to those original sinks. Compression concerns upper interfaces; it is not a preservation theorem for the physical graph's distance levels or a claim that the coupled graph has a two-sink physical realization.

2 The upper graph and its conventions

Fix $k \geq 3$ and put $L = k + 1$. A rotor stores its *next* outgoing arc. A departure uses that arc and then advances the rotor cyclically. At a binary vertex the first and second arcs are called even and odd. Thus an initially even rotor uses the even arc on its first departure. Parallel arcs, when present after compression, remain distinct positions in the rotor order.

The upper graph has absorbing boundary

$$\mathcal{B} = \{a_0, b_0, c_0\}$$

and nonsinks $a_1, \ldots, a_k$, $b_1, \ldots, b_{k-1}$, and $c_1, \ldots, c_{k-2}$. Its ordered successors are

$$\begin{aligned}
a_i &\longmapsto (a_{i-1}, a_{i+1}) && (1 \leq i < k),\\
a_k &\longmapsto (a_{k-1}, b_{k-1}),\\
b_i &\longmapsto (b_{i-1}, c_{i-1}) && (2 \leq i \leq k-1),\\
c_i &\longmapsto (b_{i-1}, c_{i-1}) && (1 \leq i \leq k-2),\\
b_1 &\longmapsto (a_k, c_0).
\end{aligned} \tag{1}$$

The a-vertices form the active line; the b- and c-vertices form the passive splitter. The arc $b_1 \to a_k$ is internal feedback.

An *external entry* places a particle at a_k or a_{k-1} and routes it until its first hit of $\mathcal{B}$. We denote these input types by E and F, respectively. Internal feedback during such an excursion is not another external entry. All boundary hits retain their names. A batch $(N, M) \in \mathbb{Z}^2_{\geq 0}$ specifies N entries of type E and M of type F; it does not specify their order.

Every vertex of the upper graph has a directed path to $\mathcal{B}$. Consequently every one-particle excursion terminates. Indeed, in an infinite walk some vertex would be visited infinitely often. Its rotor would send the particle to every successor infinitely often; iterating along a path to the boundary gives a contradiction. The same argument applies to any fixed initial phase. For finitely many particles, sequentially completing one excursion at a time gives a complete legal stabilization.

We use the complete-stabilization abelian property [7, Lemma 3.9]: for the same initial particles and rotors, complete legal routings have identical local firing counts and final rotors. One may identify the boundary vertices as a global sink for this application and retain the original labels on the incoming arcs. Local firing counts then determine how often each named output is used. This does not make the ordered word of boundary hits invariant under changes in input order.

3 Line thresholds and period-four compression

3.1 The active line

Regard the two exits of the active line as absorbing endpoints at coordinates 0 and L. The right exit from a_k enters the passive network. Write

$$R^h L^{k-h}, \qquad 0 \leq h \leq k,$$

for the next-edge state that points right at $a_1, \ldots, a_h$ and left at $a_{h+1}, \ldots, a_k$. Here L in a phase word denotes left, whereas the integer $L = k + 1$ denotes the right endpoint coordinate.

Lemma 3.1 (Line transfer)**.** *Starting from threshold h, a complete line segment entering at a_j ends at a threshold h' and has right-exit indicator e given by*

$$h' = (h + j) \bmod L, \qquad e = \left\lfloor \frac{h + j}{L} \right\rfloor. \tag{2}$$

For any finite collection of complete segments, the sum of the entry coordinates and the initial threshold equals the final threshold plus L times the number of right exits.

Proof. The last-used rotor convention of [7] is obtained by moving each next-edge rotor one position backwards. At a binary vertex the two arrows are opposite. On the line with its endpoints identified as one sink, the acyclic last-edge configurations are precisely the stated next-edge thresholds: a violation would give adjacent last-edge arrows pointing toward each other. Chip addition preserves acyclicity [7, Lemma 3.10], so a completed segment again has threshold form.

During a segment, the token coordinate plus the number of next-right rotors is invariant. A right departure increases the coordinate by one and decreases that number by one; a left departure does the reverse. On absorption this invariant reads $j + h = h' + Le$. Since $0 \le h' \le L - 1$ and $0 \le h + j < 2L$, it gives (2). Summing this identity proves the final assertion. □

3.2 The passive orbit

Temporarily cut the feedback edge $b_1 \to a_k$ and replace its destination by an absorbing symbol f. The remaining passive graph is acyclic. Start all its rotors even, insert particles only at b_{k-1}, and complete each passive traversal. Write

$$U(n) = (\mathsf{F}(n), \mathsf{B}(n), \mathsf{C}(n))$$

for the numbers of outputs f, b_0, c_0 after n traversals. These symbols denote cumulative counts, not vertices or input types.

Lemma 3.2 (Four-step reset)**.** *The complete passive rotor vector depends only on $n \bmod 4$, and*

$$U(n + 4) = U(n) + (1, 1, 2). \tag{3}$$

The repeated output word is

$$\begin{cases} (\mathsf{f}, b_0, c_0, c_0), & k \text{ odd}, \\ (\mathsf{f}, c_0, b_0, c_0), & k \text{ even}. \end{cases} \tag{4}$$

Proof. An initially even binary rotor receiving v particles sends $\lceil v/2 \rceil$ along its even arc and $\lfloor v/2 \rfloor$ along its odd arc, and ends with next phase $v \bmod 2$. After four top entries, b_{k-1} fires four times and sends two particles to each vertex of the next paired level. A pair receiving two particles at each vertex sends two particles to each vertex of the level below. Induction gives two visits to every vertex of every paired level. Every rotor therefore resets. At the bottom, b_1 sends one particle to f and one to c_0, while c_1 sends one to b_0 and one to c_0. This proves (3) and periodicity of the entire phase vector. The top rotor and its two successors distinguish the four residues: their next-phase triples are $(0, 0, 0)$, $(1, 1, 0)$, $(0, 1, 1)$ and $(1, 0, 1)$ after zero, one, two and three traversals, respectively.

The first traversal follows the b-chain and outputs f. For the first two top entries, the visit counts on the first paired level are $(1, 1)$. Descending a paired level alternates $(1, 1)$ with $(2, 0)$. Thus the bottom counts are $(1, 1)$ when k is odd and $(2, 0)$ when k is even. After subtracting the first output, the second is respectively b_0 or c_0. On the fourth traversal every departure used is the

last departure at that vertex within the four-particle batch. Since all final rotors are even, those departures are odd. This traversal follows the c-chain and exits at c_0. The total counts determine the third output. □

For later use the cumulative functions have a constant-size formula. Let $n = 4q + s$, where $0 \le s < 4$, and put

$$\beta = \begin{cases} 2, & k \text{ odd}, \\ 3, & k \text{ even}. \end{cases}$$

Then

$$\mathsf{F}(n) = q + \mathbf{1}_{s \ge 1} = \lceil n/4 \rceil, \quad \mathsf{B}(n) = q + \mathbf{1}_{s \ge \beta}, \quad \mathsf{C}(n) = n - \mathsf{F}(n) - \mathsf{B}(n). \tag{5}$$

The passive phase at residue r will mean the phase obtained from r complete traversals from the all-even state. No arbitrary passive phase is included in this notation.

Proposition 3.3 (Exact compression)**.** *On this four-phase passive orbit, replace the entire passive graph by one rotor vertex p, with cyclic successor word*

$$(a_k, b_0, c_0, c_0) \quad (k \text{ odd}), \qquad (a_k, c_0, b_0, c_0) \quad (k \text{ even}). \tag{6}$$

Replace the right arc of a_k by $a_k \to p$ and let the next rotor index at p be $r \in \{0, 1, 2, 3\}$. This preserves every sequential upper excursion, its named terminal, the active phase, and the passive residue. For batches it preserves named aggregate counts and final phases.

Proof. Each visit to p represents one complete passive traversal. By Lemma 3.2, it gives the same next destination and changes r to $r + 1 \bmod 4$. Induction over the completed traversals proves the sequential claim. Sequential routing is a complete legal stabilization; the abelian property gives the batch claim. Recovering the full original passive vector from r uses the visit-count propagation in the preceding proof. Intermediate positions inside a passive traversal are not part of the compressed interface. □

The three named sinks remain distinct in this compression. Its underlying simple undirected graph is the path $a_0 - a_1 - \cdots - a_k - p$, with two additional sink leaves b_0, c_0 at p. There are $k + 4$ vertices and $2k + 4$ arcs, including two distinct arcs from p to c_0.

4 Least stabilization and the exact decoder

Initially, a threshold h and passive residue r specify all upper rotors. For a batch (N, M) set

$$W = kN + (k - 1)M, \qquad X = h + W, \qquad f_r(t) = \mathsf{F}(r + t) - \mathsf{F}(r). \tag{7}$$

If R is the number of passive traversals, then $f_r(R)$ is the number of internal feedback entries at a_k. Summing Lemma 3.1 gives

$$R = \left\lfloor \frac{X + k f_r(R)}{L} \right\rfloor, \qquad h' = X + k f_r(R) - LR. \tag{8}$$

Flow balance alone is not enough: the actual routing selects the least nonnegative solution of the first equation.

Lemma 4.1 (Least root)**.** *The actual number of passive traversals is*

$$R = \min\{t \in \mathbb{Z}_{\ge 0} : G_r(t) > X - L\}, \qquad G_r(t) = Lt - k f_r(t). \tag{9}$$

Proof. Pause the passive region and stabilize all initial particles on the line. Then pause the line and complete all passive traversals, storing feedback particles at a_k. Repeat these waves. The cumulative top counts obey

$$R_0 = 0, \qquad R_{q+1} = \left\lfloor \frac{X + k f_r(R_q)}{L} \right\rfloor .$$

The map is nondecreasing. Since $f_r(t) \le t$, the integer X is a supersolution. The sequence therefore stabilizes at the least nonnegative fixed point. Its waves form a complete legal routing, so the abelian property identifies its counts with those of complete sequential excursions. This is a concrete instance of least action [2].

Each increment $G_r(t+1) - G_r(t)$ is L or 1, according as the next passive output is a terminal or feedback. Thus G_r is strictly increasing. At the first crossing of $X - L$, the preceding value is at most $X - L$ and the increment is at most L. Hence

$$0 \le X - G_r(R) \le L - 1 = k.$$

If $R = 0$, the same conclusion follows from $G_r(0) = 0 > X - L$ and $X \ge 0$. These inequalities are exactly (8). Conversely every fixed point has $G_r(t) > X - L$, so none precedes this crossing. □

Definition 4.2. A threshold–residue pair is *canonical* if

$$r \in \{0, 2, 3\}, \quad 0 \le h \le k, \qquad \text{or} \qquad r = 1, \quad h = k. \tag{10}$$

Lemma 4.3. *Canonical states contain the all-even state and are closed under complete external entries and batches.*

Proof. The all-even state is $(h, r) = (0, 0)$. Suppose a stabilization has $R > 0$ and final passive residue $r + R \equiv 1 \pmod 4$. Its last passive output is feedback, so the last increment of G_r is 1. The first-crossing condition in (9) forces $G_r(R) = X - L + 1$, giving $h' = k$. If $R = 0$, a canonical initial state with $r = 1$ has $h = k$; by (8), it can retain $R = 0$ only if $W = 0$. All other final residues allow any threshold. □

Put $P = 3k + 4$ and assign a canonical state the coordinate

$$z = Lr - k\lceil r/4 \rceil + h. \tag{11}$$

The following table shows that these are exactly the integers $0 \le z < P$.

Passive residue	Allowed threshold	Coordinate
0	$0 \le h \le k$	h
1	$h = k$	$k + 1$
2	$0 \le h \le k$	$k + 2 + h$
3	$0 \le h \le k$	$2k + 3 + h$

To decode an arbitrary integer $y \ge 0$, write $y = qP + t$, $0 \le t < P$, and use

range of t	h^*	n
$0 \le t \le k$	t	$4q$
$t = k + 1$	k	$4q + 1$
$k + 2 \le t \le 2k + 2$	$t - k - 2$	$4q + 2$
$2k + 3 \le t < P$	$t - 2k - 3$	$4q + 3$.

(12)

This is the unique representation

$$y = Ln - k\lceil n/4 \rceil + h^* \tag{13}$$

with $0 \le h^* \le k$ and $h^* = k$ if $n \equiv 1 \pmod 4$. Indeed, each block of four successive values of n gives the four disjoint intervals or singleton in the table; increasing n by four adds P.

Theorem 4.4 (Two-entry decoder)**.** *Let z encode a canonical state (h, r), and let $(N, M) \in \mathbb{Z}^2_{\ge 0}$. Decode $y = z + kN + (k-1)M$ using (12), obtaining (h^*, n). Then the exact final state and internal counts are*

$$z' = y \bmod P, \qquad (h', r') = (h^*, n \bmod 4), \qquad R = n - r, \qquad f = \lceil n/4 \rceil - \lceil r/4 \rceil . \tag{14}$$

In the order a_0, b_0, c_0, the named output vector is

$$C_z(N, M) = \big(N + M + f - R,\ \mathsf{B}(n) - \mathsf{B}(r),\ \mathsf{C}(n) - \mathsf{C}(r)\big). \tag{15}$$

Its entries are nonnegative integers with sum $N + M$.

Proof. By (11) and (13),

$$X - G_r(n - r) = h^*.$$

The decoder is monotone in its passive count, so $y \ge z$ implies $n \ge r$. If $R = n - r = 0$, this is already the first crossing of $X - L$. If $R > 0$ and the last increment of G_r is L, then $G_r(R-1) = X - h^* - L \le X - L$. If that increment is 1, then $n \equiv 1 \pmod 4$ and the decoder requires $h^* = k = L - 1$, again giving $G_r(R-1) = X - L$. In either case, $G_r(R) = X - h^* > X - L$, so Lemma 4.1 selects precisely this R. It also gives $h' = h^*$. The passive phase and feedback count now follow from Lemma 3.2.

There are $N + M + f$ complete line segments, of which R exit into the passive network. The rest end at a_0. The other two named counts are the passive prefix differences. Since $\mathsf{F}(n) + \mathsf{B}(n) + \mathsf{C}(n) = n$, their total is $N + M$. They count the complete legal routing just constructed, proving nonnegativity as well as the formulas. □

5 Composition, order, and bit complexity

Write $T_v(z) = z + kN + (k-1)M \bmod P$ for $v = (N, M)$. The output count is a state-dependent additive cocycle:

Corollary 5.1. *For $v, w \in \mathbb{Z}^2_{\ge 0}$,*

$$T_w(T_v(z)) = T_{v+w}(z), \qquad C_z(v) + C_{T_v(z)}(w) = C_z(v + w). \tag{16}$$

Proof. The first identity is modular addition. Routing v completely and then w is a complete legal routing of the combined batch, proving the second identity by abelianness. It can also be read directly from the decoder: removing a whole four-traversal block changes U by $(1, 1, 2)$ and the unbounded coordinate by P. Consecutive prefix differences therefore telescope, as do the internal traversal and feedback counts. □

For one entry at a_j, the theorem gives exactly one named terminal and updates z by j. Thus an ordered input word $j_1, \dots, j_s$ can be decoded at the prefix states

$$z_i = \left(z + \sum_{q<i} j_q \right) \bmod P.$$

The input multiplicities do not determine the ordered terminal word. For example, take $k = 3$ and $z = 2$. The inputs $(3, 2)$ produce the terminal word (b_0, a_0), whereas $(2, 3)$ produce (a_0, b_0). Both have the same aggregate count and final state 7. These are two choices of interface input order, with no assertion that they are prefixes of a common physical source trajectory.

Proposition 5.2 (Direct bit bound)**.** *Let*

$$b = 1 + \lceil \log_2(N + M + 1) \rceil, \qquad A = b + \lceil \log_2(k + 1) \rceil + 1.$$

For the explicit $O(k)$-vertex graph, the decoder computes all named batch counts, the internal feedback count, and the entire final rotor vector in $O(A^2 + kA)$ bit operations using elementary integer arithmetic. The scalar state itself uses $\lceil \log_2(3k + 4) \rceil$ bits.

Proof. All integers used in (7)–(15) have $O(A)$ bits. There are a constant number of additions, multiplications, divisions and comparisons, costing $O(A^2)$ by elementary multiplication and long division. Formula (5) evaluates the named passive counts with a constant number of further arithmetic operations. Alternatively, downward propagation through the passive graph takes $O(k)$ additions and halvings of $O(A)$-bit integers, costing $O(kA)$.

The threshold materializes the k active bits. For the passive vector, use $r' = n \bmod 4$ and propagate its at most three top inputs through the acyclic splitter. This takes $O(k)$ work in the fixed vertex order. The stated conservative bound covers either count evaluation and the full phase output. □

The estimate includes the binary lengths of N and M. It does not use a table whose size grows exponentially with k, and it does not materialize an ordered output word of length $N + M$. Requesting such a word entails its own output-length cost. The estimate also treats k through an explicit graph; it is not a bound for a different, succinct graph-input model.

6 An exact two-copy cyclic coupling

Take two copies, A and B, of the upper interface, with the same $k \geq 3$ and arbitrary canonical initial coordinates $z_A, z_B \in \{0, \ldots, P-1\}$. Replace the terminal edge to b_0 in copy A by an edge to a_k in copy B, and symmetrically replace the b_0 edge in B by one to a_k in A. Inject $M_A, M_B \in \mathbb{Z}_{\geq 0}$ external particles at a_{k-1} in their respective copies. The vertices $a_0(A), c_0(A), a_0(B), c_0(B)$ remain distinct absorbing outputs. We use b_0 only as a *local cut label* for the cross transfer. In particular, the coupled graph is not the physical two-sink graph of Section 8.

Every nonsink of either the compressed or the original coupled upper graph has a directed path to one of the four absorbing outputs: the line can descend to a_0, and the passive splitter has a path to c_0. The finite-path termination argument of Section 2 therefore gives a complete legal routing of every finite batch. Abelianness fixes its firing counts, final rotors, and four absorbing counts. The compression of Proposition 3.3 applies copy by copy: each local traversal still starts at one of the original two entry types, and completing a local batch at its cut outputs preserves canonicality by Lemma 4.3.

Write $b_z(N, M)$ for the b_0 coordinate of $C_z(N, M)$ in Theorem 4.4. This is a local transfer count, not a global absorbing count. Set

$$F(x) = b_{z_B}(b_{z_A}(x, M_A), M_B), \qquad U = M_A + M_B + 10. \tag{17}$$

Lemma 6.1 (Cross-count search bracket)**.** *For fixed canonical z and fixed M, the function $N \mapsto b_z(N, M)$ is nondecreasing and has adjacent increments in $\{0, 1\}$. Hence F has the same properties, $F(x) - x$ is nonincreasing, and $F(U) < U$. Its least nonnegative fixed point is*

$$x_* = \min\{x \in \{0, \dots, U\} : F(x) \le x\}. \tag{18}$$

Proof. Route a local (N, M) batch completely to the three cut outputs and then append one particle at a_k. That additional particle has exactly one cut output. Complete-stabilization abelianness identifies the result with the $(N+1, M)$ batch, so its b_0 count increases by zero or one. Composition gives the same adjacent-increment property for F.

In the decoder $y = z + kN + (k-1)M = qP + t$, the passive b_0 count is $\mathsf{B}(n) - \mathsf{B}(r) \le q + 1$. Since $0 \le z < P$ and $k/P, (k-1)/P < 1/3$, this implies the strict bound

$$b_z(N, M) < (N + M)/3 + 2. \tag{19}$$

Applying it twice gives $F(x) < (x + M_A)/9 + M_B/3 + 8/3$. Substitution of $U = M_A + M_B + 10$ shows that the right-hand side is strictly below U. Thus the set in (18) is nonempty. Because $F(0) \ge 0$ and $F(x) - x$ can decrease by at most one at an integer step, its first nonpositive value is zero. Consequently $F(x_*) = x_*$, and every other nonnegative fixed point is at least x_*. □

Theorem 6.2 (Two-copy cyclic decoder)**.** *For the exact coupling above, the physically realized number of $B \to A$ transfers is $x_{BA} = x_*$ from* (18)*; the number of $A \to B$ transfers is*

$$x_{AB} = b_{z_A}(x_{BA}, M_A). \tag{20}$$

Evaluate the source decoder twice,

$$C^A = C_{z_A}(x_{BA}, M_A), \qquad C^B = C_{z_B}(x_{AB}, M_B). \tag{21}$$

The four global absorbing counts, in the order $(a_0(A), c_0(A), a_0(B), c_0(B))$, are $(C^A_{a_0}, C^A_{c_0}, C^B_{a_0}, C^B_{c_0})$. The two final threshold–residue pairs are precisely the (h', r') outputs of the respective applications of Theorem 4.4; Lemma 3.2 recovers both full passive rotor vectors. The four absorbing counts sum to $M_A + M_B$.

With $T = 1 + \lceil \log_2(M_A + M_B + 11) \rceil + \lceil \log_2(k + 1) \rceil$, these quantities, including the explicit $O(k)$ final phase vectors, are computable in $O(\log(U + 1)(T^2 + kT))$ elementary bit operations.

Proof. First complete A's external batch at its three local cut outputs, holding its $b_0(A)$ particles at $a_k(B)$. Complete B's external batch together with those held particles, now holding its $b_0(B)$ particles at $a_k(A)$. Subsequently alternate local complete routings of newly received cross particles, retaining the outgoing cross particles at the other copy's entrance. These are legal waves in the coupled graph. After a cumulative $B \to A$ count x, local abelianness gives exactly $b_{z_A}(x, M_A)$ cumulative $A \to B$ transfers; after the corresponding B wave the cumulative $B \to A$ count is $F(x)$. Thus the cumulative counts follow iteration of F from zero.

The iteration is nondecreasing. It remains at most U, since $F(U) < U$ and F is nondecreasing. Being integral, it stabilizes at a fixed point. By monotonicity, every fixed point bounds every iterate from below, so the stable value is the least fixed point x_*. At stabilization no cross particle is pending, and the legal waves form a complete routing. Complete-routing abelianness identifies their firing counts and final rotors with those of any other legal order; the least algebraic solution is therefore the physical one. This is also consistent with the least-action formulation of Bond and Levine [2, Theorem 4.9], but the legal-wave argument establishes the specific scalar selection here.

The local formulas now give (20) and (21). Their b_0 coordinates equal x_{AB}, x_{BA} and are transfers; the other four coordinates count the absorbing outputs. Summing the local flow balances gives $(x_{BA}+M_A)+(x_{AB}+M_B)-x_{AB}-x_{BA}=M_A+M_B$. The final phases follow from the source decoder and exact compression.

Finally, the predicate $F(x)\le x$ is monotone on $[0,U]$ by Lemma 6.1. Binary search finds its first true argument in $O(\log(U+1))$ evaluations. Each uses at most two single-copy decodes, each costing $O(T^2+kT)$ by Proposition 5.2; the two final decodes obey the same bound. The bit length of each intermediate count is $O(T)$ by (19) and the bracket. This includes the cost of writing both explicit final rotor vectors, but no chip-by-chip trajectory or ordered terminal word is produced. □

The word *least* in Theorem 6.2 cannot be omitted. For $k=3$, $z_A=z_B=2$ and $M_A=M_B=0$, there is no particle to route, so the physical cross counts are $(x_{AB},x_{BA})=(0,0)$. Nevertheless $b_2(1,0)=1$, making $(1,1)$ another algebraic solution of the two cross-count equations. It is a self-supporting phantom cycle, not a legal routing from the empty input. For a positive example, $k=3$, $z_A=z_B=0$ and $M_A=M_B=3$ give $(x_{AB},x_{BA})=(1,1)$, two a_0 and one c_0 absorption per copy, and final local phases $(h',r')=(0,3)$ in both copies.

The accompanying `check_coupling.py` compares the scalar decoder with direct rotor routing on all 7,620 tested canonical compressed cases for $3\le k\le 6$, and on 324 all-even cases in the uncompressed two-copy $a/b/c$ graph. It also checks both examples above. These finite diagnostics test the paper-to-code correspondence and do not replace the proof for arbitrary k and binary batches.

7 Classical group structure and known algorithmic coverage

7.1 The sandpile calculation

For the algebraic comparison in this subsection only, identify a_0, b_0, c_0 as a single sink, retaining parallel arcs and rotor positions. The compressed graph is finite and every nonsink reaches this global sink. The sandpile group presentation has generators $[a_1],\dots,[a_k],[p]$ and relations

$$\begin{aligned} 2[a_i] &= [a_{i-1}]+[a_{i+1}] && (1\le i<k),\\ 2[a_k] &= [a_{k-1}]+[p], && \\ 4[p] &= [a_k], \qquad [a_0]=0. && \end{aligned} \tag{22}$$

Writing $g=[a_1]$, successive elimination gives

$$[a_i]=ig, \qquad [p]=(k+1)g, \qquad (3k+4)g=0.$$

There are no additional relations: the assignment $[a_i]\mapsto i$ and $[p]\mapsto k+1$ defines a homomorphism onto $\mathbb{Z}/(3k+4)\mathbb{Z}$ respecting (22). Thus the group is exactly cyclic of order $P=3k+4$.

The last-edge configuration is obtained by taking the predecessor of each next rotor position, including one step backwards in the four-cycle at p. Acyclicity on the line requires a threshold. The last arc at p is feedback exactly when its next index is $r=1$. In that case, $h<k$ would make the last arc at a_k point to p, producing a two-cycle. The condition $h=k$ removes this cycle. When $r\ne 1$, the last arc at p points to a sink and every threshold is acyclic. Consequently the acyclic last-edge states of the *compressed* graph are exactly the $3(k+1)+1$ canonical pairs.

The classical free and transitive sandpile action on acyclic rotor configurations, together with its rotor-group identification [7, Definition 3.11, Lemmas 3.12–3.19], now explains the cyclic state structure. With the all-even state as origin, chip addition at a_j acts by j in $\mathbb{Z}/P\mathbb{Z}$. Theorem 4.4

supplies its explicit coordinate and named-output formula. The original passive vectors admitted in that theorem form a restricted orbit in the uncompressed graph; they are not being identified with all its recurrent rotor configurations.

7.2 The tree-like Multi-Arrival theorem

For algorithmic evaluation keep the three sinks distinct. The compressed graph is tree-like: its underlying simple graph is the stem with two leaves described after Proposition 3.3. The nonsinks form a bidirected path, each sink has one in-neighbor, and every nonsink can reach a sink. The two arcs to c_0 are permitted parallel arcs; both orders in (6) are permitted cyclic orders.

Ghorbani, Hoff and Mnich [4, Theorem 1, p. 39:12] compute Multi-Arrival on tree-like rotor multigraphs in polynomial time when the contracted height is bounded, and return a full routing. Their model allows integer particle inputs and multiple named sinks [4, pp. 39:3–39:5]. The present tree has just one vertex of underlying degree greater than two. The source's parameter bound in terms of this number gives contracted height at most two, uniformly in our family parameter k [4, p. 39:11]. Apply the theorem to

$$\sigma = N\mathbf{1}_{a_k} + M\mathbf{1}_{a_{k-1}}.$$

The computed routing supplies each named sink count. The line firing parities determine the final active rotors; the total firing count R at p gives $r' = r + R \bmod 4$ and the feedback count $f_r(R)$. The passive vector is then expanded using Lemma 3.2.

This supplies established polynomial-time coverage after the exact compression. The source's displayed graph-size operation bound is not a literal bit bound independent of the magnitudes of N, M; their encoding must be included in a runtime comparison. Proposition 5.2 is the direct bit estimate for this decoder. No comparison establishing that this decoder is asymptotically faster is intended.

The coupling in Section 6 falls outside that particular tree-like hypothesis: in the underlying simple undirected graph it contains the cycle $a_k(A)$–$p(A)$–$a_k(B)$–$p(B)$–$a_k(A)$. This is a statement about applicability of Theorem 1 of Ghorbani, Hoff and Mnich, not a complexity separation. The single-copy theorem remains applicable to each local call of our coupling decoder. Other general methods for multi-token switching graphs are not ruled out by this four-cycle. Haslebacher's G-ARRIVAL framework already treats many-token terminal counts and gives recursive binary-search algorithms [5, Theorems 1 and 10]. The present result is the explicit polynomial-bit cross-count, phase-and-output formula for this specified cyclic wiring; no general complexity improvement is claimed.

Earlier homogeneous path results also provide useful context. Theorem 5 of Auger, Coucheney, Duhaze and Etse [1] assumes fixed arc multiplicities along the path. If b_0, c_0 are identified here, the line vertices have multiplicities $(1, 1)$ while p has one return arc and three exit arcs. Thus that theorem is not a direct hypothesis match. The same preprint proposes vertex-dependent multiplicities and different rotor orders as further directions [1, p. 28]; the later tree-like theorem already provides the relevant tractability coverage.

8 A physical two-sink realization

Return to the binary upper graph (1). Put $m = k + 4$, add original sinks $0, 1$ and lower vertices $\ell_1, \ldots, \ell_{m-1}$, and make the three boundary vertices nonsinks with successors

$$\begin{aligned} \ell_1 &\longmapsto (0, 1), \\ \ell_j &\longmapsto (a_k, \ell_{j-1}) && (2 \le j \le m-1), \\ t &\longmapsto (a_{k-1}, \ell_{m-1}) && (t \in \mathcal{B}). \end{aligned} \tag{23}$$

There is one source particle at a_k, and every rotor starts even. The lower chain uses the familiar rotor-counter mechanism; see [8, Section 2.1 and Figure 1] for its classical context.

Theorem 8.1 (Physical realization)**.** *For every $k \ge 3$, the graph* (1), (23) *has $4k + 3$ nonsinks, two original sinks, and distinct successors at every nonsink. Every vertex is graph-reachable from the source and reaches an original sink. Every shortest-distance level has size at most three. The all-even source run terminates at* 0*, uses both external upper entry vertices, and traverses $b_1 \to a_k$ during its second upper excursion.*

Proof. There are $3k-3$ upper nonsinks, three boundary vertices, and $m-1 = k+3$ lower vertices. Their displayed successors are distinct. Source reachability follows by descending the active line or the splitter and then using the boundary and lower edges; in particular b_0 remains reachable through c_1 because $k \ge 3$. Every vertex reaches an original sink by descending, using $b_1 \to c_0$ when necessary.

Define

$$d(0) = d(1) = 0, \quad d(\ell_j) = j, \quad d(t) = m \ (t \in \mathcal{B}), \quad d(v_i) = m + i \tag{24}$$

for every upper vertex $v_i \in \{a_i, b_i, c_i\}$ that exists. Every directed edge decreases d by at most one. At every nonsink there is a successor decreasing it by exactly one: for b_1 choose c_0, and for a boundary or lower vertex choose its descending successor. Thus a path of length $d(v)$ reaches an original sink, and no shorter path can do so. These are the true distances, not distances to a cut boundary. Levels below m have one vertex, level m has three, and upper levels have at most three; level zero has two sinks.

The finite-path argument from Section 2 proves termination of the rotor walk at an original sink. Its first visit to ℓ_1 uses the even edge to 0, proving the asserted terminal without any upper batch calculation.

The first upper excursion starts at a_k, descends to a_0, and leaves $(h, r) = (k, 0)$. The first departure of a_0 then sends the particle to a_{k-1}, so both external ports occur. From this state the line transfer at $j = k - 1$ has $h + j = 2k - 1 \ge L$, hence exits at the top. The initially even passive network follows its b-chain and traverses $b_1 \to a_k$. This is internal feedback within the second upper excursion. The line threshold after the first top exit is $k - 2$, and its feedback entry has $(k - 2) + k \ge L$ for $k \ge 3$, so another passive traversal follows before that excursion ends. □

This feedback is part of a strongly connected upper region. For $k \ge 4$ both children of b_{k-1} can reach b_1, and hence a_k, while a_k reaches both children. Thus branching occurs within the strongly connected component. For $k = 3$ the cycle a_3, b_2, b_1, a_3 is present alongside the bidirected active line. The uncompressed physical structure is therefore not merely the tree-like graph used for the algorithmic citation.

Proposition 8.2 (External counts in the source run)**.** *Let $H = 2^{m-2}$. If N, M count the external entries at a_k, a_{k-1} in the complete all-even source run, then*

$$N = H, \qquad H \le M \le H + 2. \tag{25}$$

The named upper return counts equal $C_0(N, M)$.

Proof. Before the first visit to ℓ_1, each ℓ_j with $j \geq 2$ is visited exactly 2^{j-1} times. To see this, ℓ_2 must depart downward once, requiring its second visit. For each higher j, only its odd departures reach ℓ_{j-1}; the last such required departure occurs on its 2^{j-1}st visit. Any later visit to that vertex would be after absorption and therefore cannot occur. In particular exactly H visits enter ℓ_{m-1}. Its lower-chain upper departures give

$$N = 1 + \sum_{j=2}^{m-1} 2^{j-2} = H,$$

where the initial source particle contributes one.

Let T_t count upper returns at $t \in \mathcal{B}$. Each boundary vertex alternates between an entry at a_{k-1} and a descent to ℓ_{m-1}. Hence $\sum_t \lfloor T_t/2 \rfloor = H$. If o is the number of odd T_t, then

$$N + M = \sum_t T_t = 2H + o.$$

The last boundary departure is descending and leaves its own count even, so $0 \leq o \leq 2$. This proves (25). The actual sequence of upper excursions starts from the all-even canonical state; Theorem 4.4 and complete-stabilization abelianness identify its named aggregate returns with $C_0(N, M)$. □

The external injection at a_k consists of the original source and even departures from the lower vertices. The external injection at a_{k-1} consists of even departures from the boundary vertices. The $b_1 \to a_k$ traversals lie inside the upper cut and are counted separately as f. Thus the physical construction uses the same two-input interpretation as the abstract interface. The proposition does not require a new source at each feedback traversal.

9 Scope of the worked model

The exact decoder uses a threshold line, one feedback destination, and the four-phase orbit of the specified homogeneous splitter. It covers every canonical upper state and binary batch on that interface. Arbitrary passive initial configurations, other splitter layouts, or several feedback destinations are not covered by the stated coordinate. The coordinate is an exact sufficient encoding; no minimal-transducer assertion is needed.

The physical family provides a concrete width-at-most-three realization. It does not settle the general bounded-width ARRIVAL problem. In particular, the concluding direction in Gärtner, Haslebacher and Hoang [3, Section 6] asks about graphs whose every nonempty distance level has the *same* fixed size $r > 2$. The lower levels here have size one, so this family does not match that precise domain. Its original source terminal is already forced to 0 by the first departure from ℓ_1; its purpose is to realize the upper-interface law, not to supply a difficult new winner decision. Nor is there an all-strategy stopping stochastic-game conclusion: if lower vertices were controlled and always selected their upper successors, the original sinks could be avoided.

Classical rotor-group structure and the existing tree-like algorithm account for the single-copy cyclic action and polynomial batch computability after compression. The two-copy theorem gives an exact specialized computation of physical cross counts, final phases, and four absorbing counts. Neither the two-copy upper graph nor its binary-batch inputs are the physical two-sink single-source construction of Section 8. The FUN 2026 ladder algorithm [3, Theorem 10] concerns one-token ARRIVAL on graphs whose nonempty sink-distance levels have exactly two vertices, apart from a

possible final one-vertex level. Our physical lower levels have size one and some upper levels have size three; its source terminal is already forced to 0. The two-copy upper graph is a different four-output batch model. These scope differences do not establish novelty or hardness. The literature comparison is bounded; in particular, the mathematical content of Hoang's thesis [6] is not used to establish a priority comparison here.

References

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